Homework 3: Regression Review and Prediction Intervals

ISYE 4031 - Summer 2026 - Problems Only Version

Submission Instructions

Include your full name, Georgia Tech email, and GTID in your submitted work. You may complete the homework using JupyterLite, a local Python or R environment, spreadsheet software, handwritten work where appropriate, or another approved environment. Your submission should clearly answer each question and show enough reasoning, formulas, code output, or supporting work to be graded.

Required Data Files

- county_bank_branches.csv
- ozone_trends.csv

Problem 1: Bank Branch Access in New Jersey Counties

The file `county_bank_branches.csv` contains 21 New Jersey counties. The response is the number of residents per bank branch. The predictor is the percentage of minority population in the county.

Answer parts (a) through (h). Use complete sentences for interpretation questions.

1(a) Create a scatter plot

Make a scatter plot of residents per bank branch versus minority population percentage. Briefly describe the direction and strength of the relationship.

1(b) Fit the simple linear regression model

Fit a model of `ResidentsPerBranch` as a function of `MinorityPercent`. Write the fitted equation.

1(c) Interpret the estimated intercept and slope

Interpret both coefficients in context. If the intercept is not practically meaningful, say why.

1(d) Test whether the coefficients are significant

Use two-sided tests at $\alpha = 0.05$ for both the intercept and the slope. For each coefficient, state the hypotheses, test statistic or p-value, conclusion, and contextual interpretation.

1(e) Predict residents per bank branch when minority population is 50%

Report the fitted value.

1(f) Construct a 95% confidence interval for the mean response at 50%

Interpret the interval in context.

1(g) Construct a 95% prediction interval for a future county at 50%

Interpret the interval in context.

1(h) Compare the two intervals

Explain why the prediction interval should be wider than the confidence interval for the mean response.

Problem 2: Ozone Warning Days and Meteorological Index

The file `ozone_trends.csv` contains yearly ozone warning days and a meteorological index. Answer parts (a) through (f).

2(a) Create a scatter plot

Plot ozone warning days against meteorological index and describe the relationship.

2(b) Fit the regression equation

Fit `OzoneDays` as a function of `MeteorologicalIndex`. Report the fitted equation.

2(c) Test whether the regression is significant

Use $\alpha = 0.05$. State the hypotheses, p-value, conclusion, and contextual interpretation.

2(d) Predict ozone warning days when the meteorological index is 17

Report the fitted value.

2(e) Construct a 95% confidence interval for the mean response at index 17

Interpret the interval in context.

2(f) Construct a 95% prediction interval at index 17

Interpret the interval and compare it with the confidence interval.

AI Usage Report

Your submission must include an AI Usage Report following the course AI policy. If you used AI tools, include the tool(s), what you asked them to help with, what output you accepted, changed, or rejected, and how you verified the final work. If prompts or screenshots materially affected your submission, include copied prompt text or screenshots in the report when appropriate. If no AI tools were used, include this statement: “No AI tools were used for this submission. You may include this report at the end of your submission or attach it as a separate file.